

DAnCing body, Speaking Hands (DASH): Sign Dance Generation System with Deep Learning

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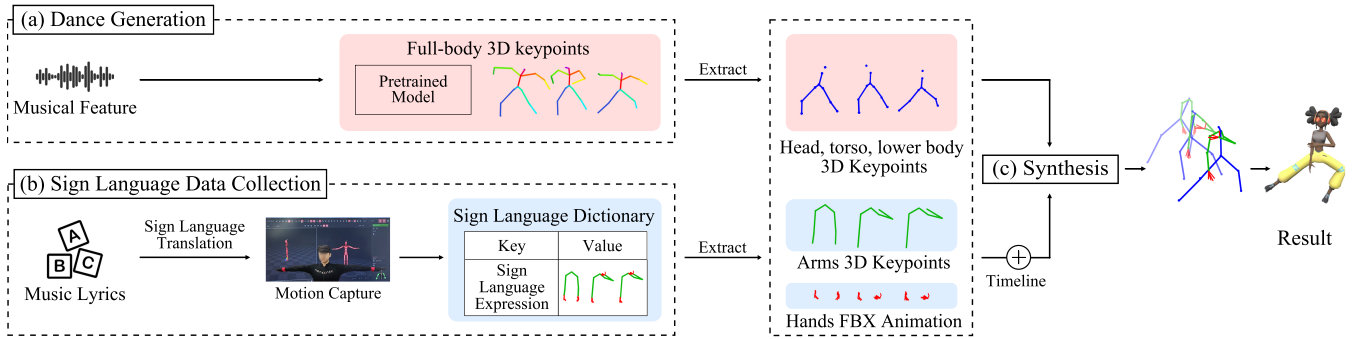


Figure 1: Overall architecture of the DAnCing body, Speaking Hands (DASH)

KEYWORDS

visualization, music, dance generation, disability, accessibility

ACM Reference Format:

Eunhee Kim, Taehwa Park, Jaeyoung Moon, Wonsang You, Taegwan Ha, and Kyung-Joong Kim. 2023. DAnCing body, Speaking Hands (DASH): Sign Dance Generation System with Deep Learning. In *Special Interest Group on Computer Graphics and Interactive Techniques Conference Posters (SIGGRAPH '23 Posters)*, August 06–10, 2023, Los Angeles, CA, USA. ACM, New York, NY, USA, 2 pages. <https://doi.org/10.1145/3588028.3603687>

1 INTRODUCTION

Disability should not be a barrier to experiencing content. However, the deaf and hard-of-hearing (DHH) communities have had limited listening experiences, which hinders musical appreciation. Therefore, various approaches to modality transitioning from the auditory to other senses including the visual have been proposed for the DHH [Petry et al. 2016].

Among various visualizers such as color spectrum and waveforms [Petry et al. 2016], dance has been proposed as a promising method for intuitively expressing musical features [Siyao et al.

2022]. However, dance alone is insufficient for fully representing music for the DHH because it excludes lyrics. In this context, the previous approach [Iwamoto et al. 2019] showed the possibility of using sign language for representing lyrics with dance. It is used to generate sign dance from the collected full-body sign dance database.

We propose Dancing body, Speaking Hands (DASH) that performs sign dance with synthesizing dance from deep learning techniques and sign language. It is able to express diverse movements and meanings of the lyrics from a music piece, beneficial to enrich the musical experience for the DHH. To the best of our knowledge, this is the first work that applies deep learning to sign dance generation.

2 APPROACH

Figure 1 presents the overall architecture of DASH. Unlike other music visualization methods, our method delivers both musical features and lyrics intuitively to the DHH in the form of sign dance. To synchronize DASH, we compose the dance and sign language in three stages. First, a sequence of full-body dance is generated using the musical features (Fig. 1a). Second, the sign language motion capture (MoCap) data, collected in the form of a dictionary (Fig. 1b) with a motion capture system from Nansense Inc., is selected from the lyrics. Last, the synchronized sign dance is generated using Inverse Kinematics (IK) and animation techniques (Fig. 1c). The dance is visualized with a character from Mixamo [Adobe 2007].

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SIGGRAPH '23 Posters, August 06–10, 2023, Los Angeles, CA, USA

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ACM ISBN 979-8-4007-0152-8/23/08.

<https://doi.org/10.1145/3588028.3603687>



Figure 2: Comparison of Sign dancer and KSL expression

2.1 Dance Generation

We adopted a fine-tuned deep learning model from Bailando [Siyao et al. 2022] with AIST++ [Li et al. 2021] 3D dance dataset. We excluded the specific dance types such as break-dancing, as they include improper arm and hand movements that conflict with the movements for speaking sign language. The pretrained model takes musical features to generate the 3D keypoints for dance sequence.

2.2 Sign Language Data Collection

We selected 13 Korean songs of various genres, including dance, rock, and ballad. Then, a professional sign language interpreter translated lyrics into Korean Sign Language (KSL) in word units. Finally, we used the MoCap system to collect the precise trajectories of the signing motion in 3D space and recorded 284 MoCap data in a total of 34,740 frames. We established a sign language dictionary that included KSL expressions as the keys and the KSL MoCap data as the values.

2.3 The Complete Sign dancer

DASH synthesizes (Fig. 2a) body movements from the generated dance and hand gestures from the sign language dictionary. The dance movements were extracted using the head, torso, and lower body 3D keypoints from generated full-body dance data. For the signing gestures, we selected a set of KSL expressions to the given lyrics (Fig. 2b). Then, we divided the KSL MoCap data into arms and hands. We represented motions of arms using 3D keypoints, and hands as FBX animations to synthesize with the body movements.

We used Unity 3D engine to composite the DASH so that the dancer would express body movements for the dance, while speaking sign language. We controlled the head, torso, lower body, and arms using IK. It helps determine the joint angles and positions for adjusting the proper position and orientation of the hand gestures, which achieves the natural connection between 3D keypoints and FBX animations of hand gestures. Moreover, Unity Animator Component was used to animate the signing hands along the proper timeline of lyrics. As the primary purpose of the DASH hand gestures is to convey the meaning of the lyrics correctly, we adopted a strategy that would directly animate the recorded motion.

3 PILOT STUDY

We performed a pilot study to validate the effectiveness of the DASH. 11 DHH individuals participated in the pilot study (age: $M : 44.73$, $SD : 6.12$; 5 males and 6 females). Participants responded to an interview ("how did you experience the sign dancer compared to the subtitles?") after watching dancing videos; sign dance (E1) and dance with subtitles (E2).

Three participants appreciated the DASH. P3 responded that ("It was good to watch. I hope I can feel the inner [mind] sound.") Conversely, Five participants pointed out the limitation of the DASH that the sign gestures were not well demonstrated. P10 responded that ("It looked like good performance, but I rarely realized the sign gestures.") Others responded that the DASH was not too bad but also not so good.

Through the pilot study, we confirmed that the DHH hopes to feel the meaning of the music and the DASH can work for them. Also, we found that the DASH needs to be developed to denote more detailed expressions of the hand gesture.

4 CONCLUSION

We proposed the novel sign dance generation system called DASH. Our system synthesizes a sequence of dance generated from the deep learning model with sign language MoCap data, which is the first approach for sign dance generation. Pilot study showed the validity of DASH for the DHH with further improvements.

Currently, our system has limitation of controlling the detailed movements of the dance. We plan to improve IK to control detailed dynamics, which will enable the abundant expression of DASH. Our system presents the possibilities for enriching musical experiences for the DHH. We hope that our work will inspire further research into the technologies for underrepresented minorities.

ACKNOWLEDGMENTS

This research was supported by Culture, Sports and Tourism R&D Program through the Korea Creative Content Agency grant funded by the Ministry of Culture, Sports and Tourism in 2022 (Project: Development of technology to visualize music and dance for the hearing impaired to enjoy music, Project Number: R2021040048, Contribution Rate: 100%)

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